605.486 – Android Development

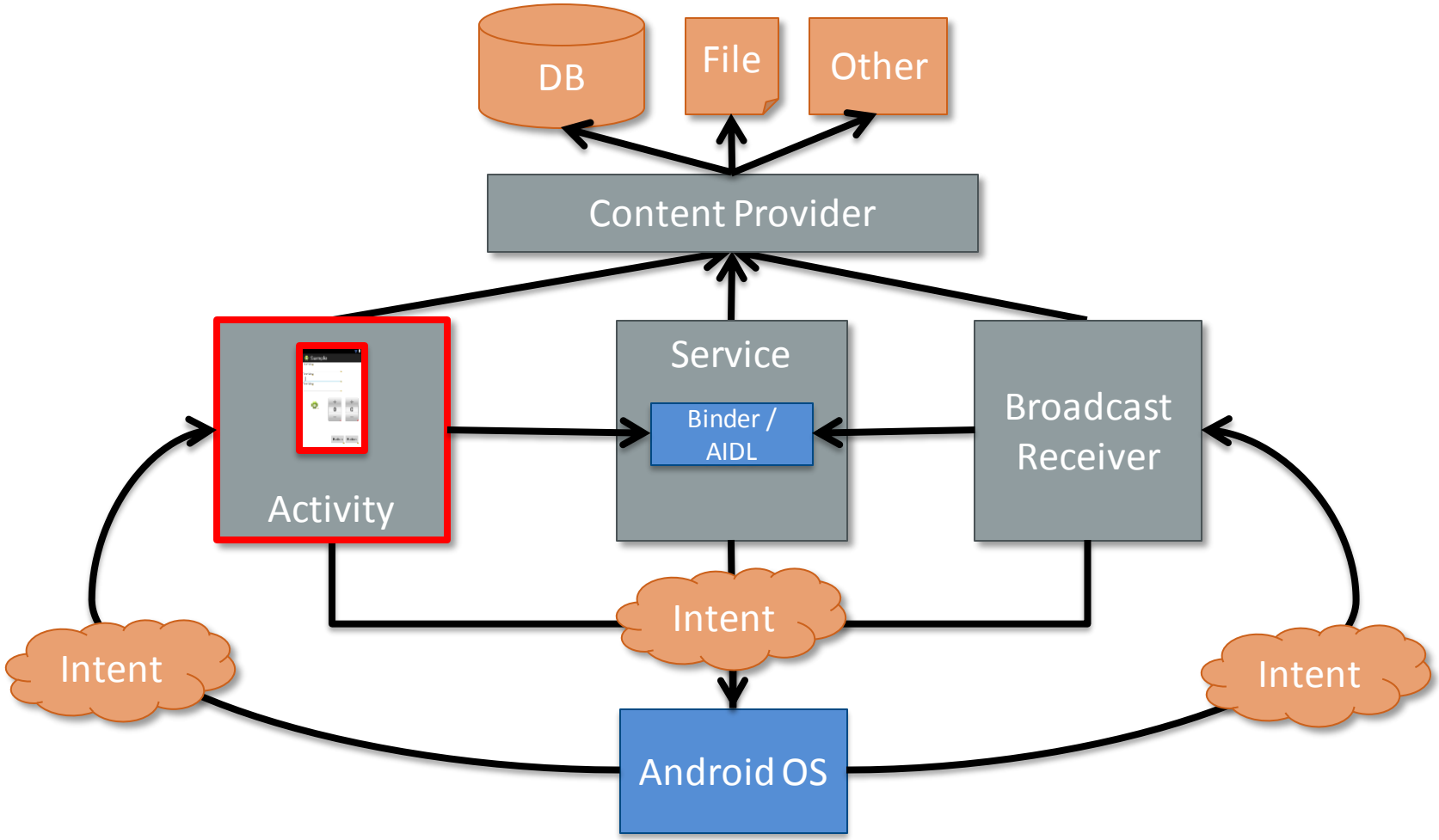
Module 2a Activities

Overview

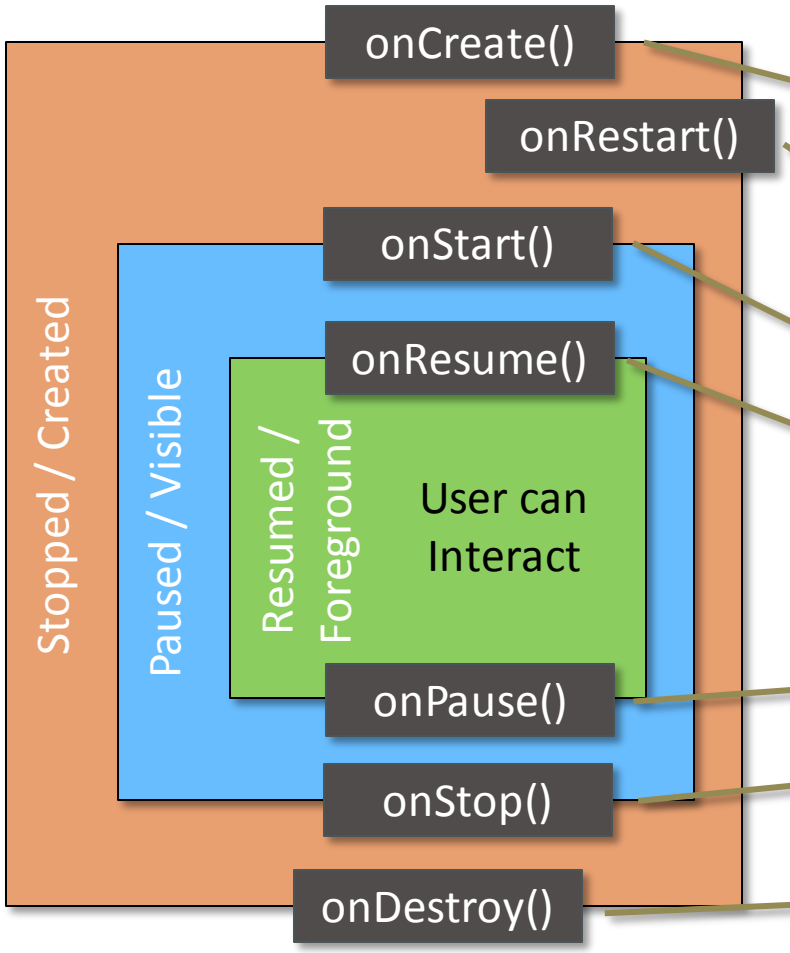
- Application Components
  - Top-level chunks of an application
- Activity
  - User interface "screen"
  - Typically acts as a single "window" or "panel"
  - Often multiple Activities in an Application

Application Components

and Typical Communication



Activity Lifecycle



Activity Triggered
Create views, fetch data
Setup view listeners, Restore saved state

User navigates back
Redo onCreate() actions if needed

About to make visible
Refresh displayed values, Bind services

About to make interactive
Start animations/sounds/service listeners

About to make non-interactive
Save data
Stop Animations/sounds/service listeners

About to make non-visible
Unbind services

About to obliterate
Final cleanup, if any

Process Death

- Paused/Stopped processes may be killed!
- onStop() / onDestroy() *may never be called*
  - Be sure anything these methods do are covered by a process being killed
    - Setting objects to null
    - Unbinding services
  - Data saving/commits should be done in onPause() or as needed

Resources and the "R" Class

- res
  - drawable-hdpi
    - ic_launcher.png
  - drawable-ldpi
  - drawable-mdpi
  - drawable-xhdpi
  - drawable-xxhdpi
  - layout
    - activity_main.xml
  - menu
    - main.xml
  - raw
    - sample.xml
  - values
    - dimens.xml
    - strings.xml
    - styles.xml
  - values-sw600dp
    - dimens.xml
  - values-sw720dp-land
    - dimens.xml
  - values-v11
    - styles.xml
  - values-v14
    - styles.xml

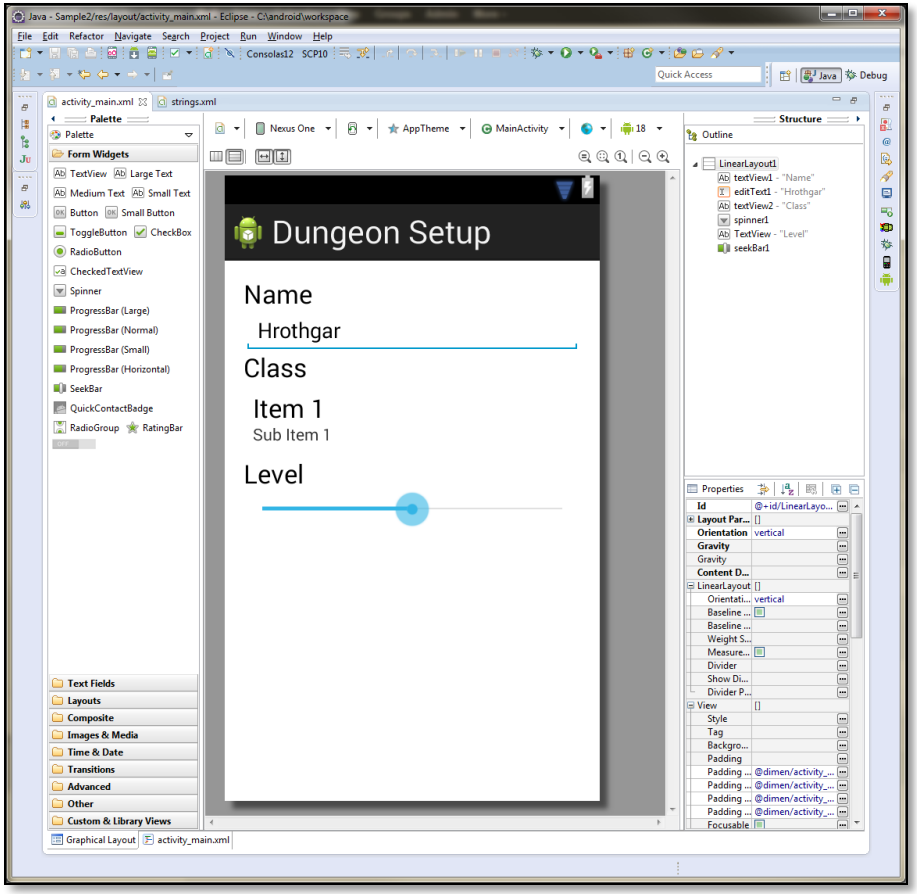
```xml
<menu xmlns:android="..." >
    <item
        android:id="@+id/action_settings"
        android:orderInCategory="100"
        android:showAsAction="never"
        android:title="@string/action_settings"/>

</menu>
```

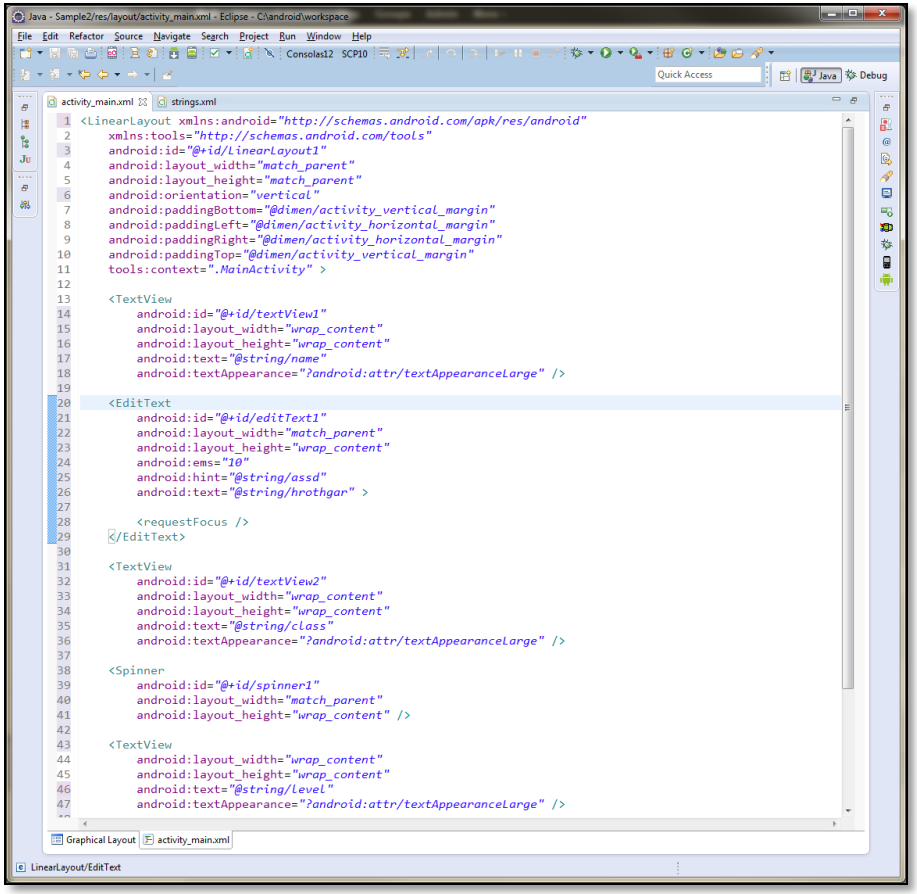
```xml
<resources>
    <string name="action_settings">
        Settings
    </string>
    <string name="app_name">
        Sample2
    </string>
    <string name="hello_world">
        Hello world!
    </string>
</resources>
```

```xml
<resources>
    <style name="AppBaseTheme"
        parent="android:Theme.Light" />
    <style name="AppTheme"
        parent="AppBaseTheme" />
</resources>
```

```java
public final class R {
    public static final class attr {
    }
    public static final class dimen {
        public static final int activity_horizontal_margin=0x7f050000;
        public static final int activity_vertical_margin=0x7f050001;
    }
    public static final class drawable {
        public static final int ic_launcher=0x7f020000;
    }
    public static final class id {
        public static final int action_settings=0x7f090000;
    }
    public static final class layout {
        public static final int activity_main=0x7f030000;
    }
    public static final class menu {
        public static final int main=0x7f080000;
    }
    public static final class raw {
        public static final int sample=0x7f040000;
    }
    public static final class string {
        public static final int action_settings=0x7f060001;
        public static final int app_name=0x7f060000;
        public static final int hello_world=0x7f060002;
    }
    public static final class style {
        public static final int AppBaseTheme=0x7f070000;
        public static final int AppTheme=0x7f070001;
    }
}
```

GUI Layout


Dungeon Setup
Name
Hrothgar
Class
Item 1
Sub Item 1
Level


activity_main.xml
strings.xml
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/LinearLayout1"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity" >
<TextView
android:id="@+id/textView1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="@string/name"
android:textAppearance="?android:attr/textAppearanceLarge" />
<EditText
android:id="@+id/editText1"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:ems="10"
android:hint="@string/assd"
android:text="@string/hrothgar" >
<requestFocus />
</EditText>
<TextView
android:id="@+id/textView2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="@string/class"
android:textAppearance="?android:attr/textAppearanceLarge" />
<Spinner
android:id="@+id/spinner1"
android:layout_width="match_parent"
android:layout_height="wrap_content" />
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="@string/level"
android:textAppearance="?android:attr/textAppearanceLarge" />
LinearLayout/EditText